

Thirulok Sundar Mohan Rasu

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Education

University of Michigan — Ann Arbor

Aug 2025 – Dec 2026

M.S., Electrical and Computer Engineering

GPA: 4.0 / 4.0

Relevant courses: *Matrix Methods for ML & Signal Processing (ECE 551)*, *Probability & Random Processes (ECE 501)*

Indian Institute of Technology (ISM) Dhanbad

Dec 2021 – May 2025

B.Tech., Electrical Engineering

GPA: 8.30 / 10

Skills

Languages: Python, Julia, MATLAB, C++

Research Areas: Scientific ML, Physics-Based Modeling, Optimization, Computer Vision, Image Segmentation, Time-Series Prediction

Tools: PyTorch, OpenCV, SciPy, ForwardDiff.jl, ROS2, GitHub, Linux, MuJoCo, Gazebo, PyBullet, Simulink

Experience

Electric Vehicle Center — University of Michigan

Oct 2025 – Present

Graduate Research Assistant (Prof. Ziyu Song)

- Built physics-based EIS modeling framework combining equivalent circuit models (SEI/CEI layers) with DRT analysis; extracted impedance parameters tracking SEI evolution from BOL to EOL via Differential Evolution optimization, achieving **1.7% mean error** across 117 EIS spectra
- Developed attention-enhanced Seq2Seq models for battery aging prediction, achieving **2–5% MAPE** on full capacity-fade trajectories from the first 30% of cycling data with robust generalization across C-rates and depths of discharge
- Designed accelerated testing protocol framework converting real-world driving profiles to battery power demands via vehicle dynamics model; optimized protocols to maximize degradation mechanism diversity for efficient lab characterization

University of Michigan — Prof. Raj Rao Nadakuditi

Jan 2026 – Present

Graduate Research Assistant

- Translating the CyScat electromagnetic scattering simulation codebase from MATLAB to Julia and Python; understanding and implementing S-matrix methods for periodic arrays of cylinders with T-matrix multiple scattering
- Studied wavefront optimization via SVD of scattering matrices — the first right singular vector of S_{11}/S_{21} identifies optimal eigenchannel wavefronts achieving near-perfect transmission through complex structures

Carnegie Mellon University

Jan 2024 – Sep 2025

Research Intern (Dr. Arun Balajee Vasudevan)

- Developed content-agnostic deepfake speech detection model trained on 1M+ real-fake speech pairs using multiple TTS systems (YourTTS, XTTS-v2, OpenVoice); model outperforms human listeners across all speech lengths
- Paper submitted to **Interspeech 2026** (under review)

Mowito

Mar 2023 – Jul 2023

Robotics Software Intern

- Developed Multi-head Mask R-CNN models for instance segmentation in warehouse environments; implemented perceptual hashing for dataset deduplication, reducing dataset size by 30%
- Built modular dataset pipeline with multiple annotation format support; developed order-fulfillment time estimation using Leiden clustering for optimized crate grouping

Selected Projects

Challenges in Multi-view 3D Scene Reconstruction (Course Project) — Analyzed failure cases of MAST3R under challenging capture conditions; collected custom dataset with varying lighting, occlusions, and viewpoints; documented systematic limitations.

Robot State Estimation: EKF & Particle Filters (GitHub) — Implemented and benchmarked EKF and Particle Filter for PR2 localization across five scenarios; analyzed accuracy vs. compute trade-offs.

Publications

T. S. Mohan Rasu, A. Suresh, A. B. Vasudevan, “Towards Content-Agnostic Deepfake Speech Detection with Multi-TTS,” submitted to *Interspeech 2026* (under review).